

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian RVs

1 Marginal PMF

1.1 Definition

Joint PMF:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Marginal PMFs:

$$p_X(x) = \sum_y p_{XY}(x, y), \quad p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea: Sum over the other variable.

1.2 From Joint PMF Table

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

1.3 Worked Example

	$y = 0$	$y = 1$
$x = 0$	$1/8$	$1/4$
$x = 1$	$3/8$	$1/4$

Step 1:

$$p_X(0) = \frac{1}{8} + \frac{1}{4} = \frac{3}{8}$$

$$p_X(1) = \frac{3}{8} + \frac{1}{4} = \frac{5}{8}$$

Step 2:

$$p_Y(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

2 Correlation

2.1 Definition

$$R_{XY} = E[XY]$$

Interpretation: Average value of the product XY .

Limitation: Not centered about means.

3 Covariance

3.1 Definition

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

3.2 Derivation

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y] \\ &= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X\mu_Y \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

3.3 Properties

- $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
- $\text{Cov}(X, X) = \text{Var}(X)$
- $\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$
- Independence $\Rightarrow \text{Cov}(X, Y) = 0$

3.4 Example

Given:

$$Y = -2X + 3$$

$$XY = -2X^2 + 3X$$

$$E[Y] = -2E[X] + 3$$

$$\text{Cov}(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

$$= -2E[X^2] + 2(E[X])^2$$

$$= -2\text{Var}(X)$$

4 Pearson Correlation Coefficient

4.1 Definition

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$-1 \leq \rho_{XY} \leq 1$$

5 Continuous Example

Given:

$$f_{XY}(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

5.1 Step 1:

$$E[X] = \frac{1}{8} \int_0^4 x^2 dx = \frac{8}{3}$$

5.2 Step 2:

$$E[Y] = \frac{1}{12} \int_1^5 y^2 dy = \frac{31}{9}$$

5.3 Step 3:

$$E[XY] = E[X]E[Y] = \frac{248}{27}$$

5.4 Step 4:

$$E[2X + 3Y] = \frac{47}{3}$$

5.5 Step 5:

$$\begin{aligned} E[X^2] &= 8 \\ \text{Var}(X) &= \frac{8}{9} \end{aligned}$$

5.6 Step 6:

$$\begin{aligned} E[Y^2] &= 13 \\ \text{Var}(Y) &= \frac{92}{81} \end{aligned}$$

5.7 Step 7:

$$\text{Cov}(X, Y) = 0$$

5.8 Step 8:

$$\rho_{XY} = 0$$

6 Joint Gaussian Random Variables

6.1 Definition

A pair (X, Y) is jointly Gaussian if the joint density is bivariate Gaussian.

6.2 PDF

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left(\frac{-1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) \right] \right)$$